



Department of Electrical and Computer
Engineering
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Design and Implementation of a 3-Stage Elevator System Using HCS12 Microcontroller

I. Project Description:

Design a **3-stage elevator system** that uses MCCNT UNDERFLOW INTERRUPT to check periodically the state of three external push buttons and three internal push buttons and update the elevator's movement accordingly. The system is equipped with a weight sensor to indicate any potential overload.

II. System Operation:

- The elevator includes :
 - Three internal push buttons for floor selection inside the elevator : PB3-PB4-PB5 connected to PORTA pins : PA2-PA3-PA4 respectively.
 - Three external push buttons , one on each floor for requests outside the elevator: PB6-PB7-PB8 connected to PORTA pins : PA5-PA6-PA7 respectively.
 - One green LED connected to PORT T pin 7 , PT7 , that will blink when the elevator is moving upward.
 - One yellow LED connected to PORT T pin 6 , PT6 , that will blink when the elevator is moving downward.
 - One red LED connected to PORT T pin 5 , PT5 , that will blink when the door opens.

- Using the **MCCNT underflow interrupt**, the system must periodically check the state of all push buttons and update the elevator's movement accordingly, either **upward** or **downward**, toward the requested floor.
- If the requested floor is greater than the current one :
 - The elevator will move up , the text 'MOVING UP' will be displayed on the LCD and the green LED will blink continuously during motion.
 - Upon arrival :
 - The green LED turns OFF .
 - The buzzer beeps for a given time .
 - The red LED blinks indicating that the door is opening.
 - The current floor number , 'F2' *for example* , is displayed on the LCD.
- If the requested floor is lower than the current one :
 - The elevator will move down, the text 'MOVING DOWN' will be displayed on the LCD and the yellow LED will blink continuously during motion.
 - Upon arrival , the same steps described above will be repeated.
- If 2 buttons are pressed at the same time and the destination floors are above the current one then the requests will be served in ascending order.
Ex: if the current floor is GND floor and the destination floors are floor1 and floor2 simultaneously , so the elevator will move up to the first floor then to the second one.
- If 2 buttons are pressed at the same time and the destination floors are below the current one then the requests will be served in descending order.
Ex: if the current floor is floor 2 and the destination floors are floor1 and GND simultaneously , so the elevator will move down to the first then to the GND floor.
- The potentiometer will act as a weight sensor:
 - If the measured weight exceeds a predefined threshold, the buzzer will beep , and the elevator door will remain open (*the red LED will remain blinking*) until the weight returns to an acceptable level.

III. Guidelines

- You have to submit: Report , PowerPoint presentation and CodeWarrior Program.
- Follow the same guidelines of the Lab reports: Same Outlines, an introduction and a conclusion, Table of contents, Table of figures,
- Explain **in details and very clearly** the report and the Code of every task(write comments).
You can use a flowchart.
- Write the issues (if any) that you have encountered.
- Insert screenshots of the breadboard connections.
- The presence of all the team members during the demo is crucial, and every team member should be ready to answer any question.
- Cheating is not tolerated , and LAU code will be strictly enforced.
(any paraphrasing of someone else's report will prevent both groups from grading)
- The deadlines (as sent in blackboard) are final and non-extendable.
- You have 10 min to demonstrate your project and test it on the board.

IV. Grading Scheme

The grading scheme will be divided as follows:

- Achieving all the tasks successfully.
- Optimization of the code.
- The report.
- The PowerPoint presentation: Every team member should be ready to answer any questions during the demo.
- The submission of all requirements on time.
